Is the galaxy RAR scale $cH_0/6$? A data-first reading of 12 X-COP cluster profiles and 13 CLASH cores

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Abstract

The radial acceleration relation (RAR) of disc galaxies has a single scale, $a_0 = (1.20 \pm 0.02 \pm 0.24)\cdot 10^{-10}$ m s^{-2}, numerically close to cH_0 (Milgrom: $cH_0/2\pi$; Verlinde's emergent-gravity construction: $cH_0/6$). Galaxy clusters show a similar low-acceleration relation with scales about ten times higher. This record looks at three measured things and then reads them with the author's framework, Mathematical Materiality (MM). The author's method is to go from the data to the framework: readings found after seeing the data are stated as such, protocols and local registrations are kept, and the checks by a second AI assistant are documented. (1) The galaxy scale. MM's two-end reading gives $cH_0/2$, 2.73 times the measured a_0. Dividing by 3 gives $a_0 = cH_0/6 = 1.09$ ($H_0 = 67.4$) or 1.18 ($H_0 = 73.0$) $\cdot 10^{-10}$ m s^{-2}, inside the measured band; in MM the factor is read as the rule used in MM 2.44 for Ω_c/Ω_b = 3√3, counted on two directions. The count, the direction of the factor and the identification with a_0 were chosen knowing the target. Approximate coherence checks: vertical density near the Sun (63.9 vs 68 ± 4 M⊙ pc^{-2}, against 107 with $cH_0/2$), the 3√3 cluster baryon fraction (unchanged), the shape of $a_0 \propto H(z)$ (compatible with free amplitude if the MUSE-DARK III bars are 1σ).

Ugo Lanciano · Independent researcher, Locri, Italy · ORCID 0009-0008-7618-9635 · October 2026. Original text and materials: CC BY 4.0. Original code: MIT licence. Third-party data are not covered by these licences and keep the conditions of their sources (`fonti/SOURCES.md`). In the working records this deposit collects premises 43-47 (folders 30-34 of the working tree); some file names carry those numbers. They belong to this record. The check added in version 2.46.1 comes from premise 48 (folder 36).

In brief

The radial acceleration relation (RAR) of disc galaxies has one scale, $a_0 = (1.20 \pm 0.02 \pm 0.24)\cdot 10^{-10}$ m s^{-2}, numerically close to cH_0 (Milgrom: $cH_0/2\pi$; Verlinde's emergent-gravity construction: $cH_0/6$). In galaxy clusters a similar low-acceleration relation holds with scales about ten times higher. This record looks at three measured things and then reads them with the author's framework, Mathematical Materiality (MM):

1. **The galaxy scale.** MM's two-end reading gives $cH_0/2 = 3.27\cdot 10^{-10}$ m s^{-2}, 2.73 times the measured a_0. Dividing by 3 gives $a_0 = cH_0/6 = 1.09\cdot 10^{-10}$ ($H_0 = 67.4$) or $1.18\cdot 10^{-10}$ ($H_0 = 73.0$) m s^{-2}, inside the measured band (section 3). In MM the factor 3 is read as the rule that MM 2.44 uses for Ω_c/Ω_b = 3√3 (√3 per direction between the visible and a hidden sector), counted on two directions. The count, the direction of the factor and the identification of the two-end threshold with the RAR parameter a_0 are choices made knowing the target.
2. **Twelve X-COP clusters.** The scale inferred point by point with the galaxy formula decreases with radius in all twelve clusters (median slope −0.97, range −1.44 to −0.61, over 0.2-1.2 R500). A scale A/r (one amplitude per cluster) fits every profile better than a constant scale. The registered test of a «history» dependence (one scale per cluster at R500 against concentration and central entropy) gives no indication (Spearman −0.09 and +0.02). In MM the A/r behaviour is read as a fixed distal end: the governing variable becomes G·M_bar(<r)/r. An ordinary explanation (NFW-like total mass with a baryon fraction rising outwards) gives the same slope (section 5).
3. **Thirteen CLASH cores.** Inside the brightest cluster galaxies (14.3-30.3 kpc) the registered test gives no indication: in 10 of 13 cores the tabulated baryonic mass exceeds the lensing mass. Exploratory: the two cluster laws of item 2, extended inwards, predict 2-15 times the lensing mass (section 6).

A fourth part records readings that fall: written with counts, the onset of the dark part is at $c^4/(4G\cdot M_U)$ (under the capacity rule and spherical-equivalent masses), and eight readings that tie it to the centre of each galaxy give scales that vary from galaxy to galaxy, against the single RAR scale (section 4).

Method, declared. The author's instruction is to go from the data to MM: a reading found after seeing the data is acceptable, the criterion is coherence. Protocols and local registrations (SHA-256 and local time, not an independent timestamp) are kept for each step; readings built knowing the target and analyses added afterwards are marked as such.

Checks by a second AI assistant (Codex, «Astra») are documented in the reports kept in `documenti/rapporti_astra/` (Italian).

Reproduction: `python scarica_dati.py` (once; downloads SPARC from CDS and the X-COP release, checks SHA-256), then `python riproduci.py` (6 modules, 63 checks of the main numerical results; reproduction and consistency checks, not physical confirmations; the download path itself is not among them).

1. What a physicist will recognise

- **The RAR** (McGaugh, Lelli & Schombert 2016), g_obs = g_bar/(1 − e^(−√(g_bar/a_0))), and the closeness of a_0 to cH_0: $a_0 \approx cH_0/2\pi$ (Milgrom); a_M = $cH_0/6$ in Verlinde (2016), with a profile-dependent law for extended systems.
- **The RAR in clusters:** CLASH $2.02\cdot10^{-9}$ (Tian et al. 2020), X-COP $1.12\cdot10^{-9}$ (Pradyumna et al. 2021), X-ray groups $0.88\cdot10^{-9}$ m s^{-2} (Gopika & Desai 2021), all obtained by imposing g_tot = √(a_0 g_bar); and the long-standing MOND mass deficit in clusters.
- **Potential-dependent modifications** of MOND in clusters (EMOND: Zhao & Famaey 2012; Hodson & Zhao 2017).
- **The splashback radius** of clusters, about 0.8-1.5 R200m, roughly 1.5-2 R200c for massive low-redshift clusters (More, Diemer & Kravtsov 2015; Bianconi et al. 2021).

2. The reading used

- L = R_s/r («stacked cubes»: the mass within r relative to the maximum that radius can hold) and S = r/R_H («cubes side by side»: the space up to r relative to the Hubble radius). They are equal at r_t = √(R_s R_H); for g_N = GM/r², L/S = g_N/(cH_0/2) exactly.
- In the author's book NUMERI the ratio g_dark/g_bar is the sum of reading steps 1/(e^√(L/S) − 1); the sum has the functional form of the RAR. The step factor is a choice; the RAR form is adopted from the data. Identifying the two-end threshold with the parameter a_0 of that formula is a further choice: other definitions of the observed onset give other numbers (for example, dark part equal to the baryonic part at g_bar = (ln 2)² a_0).
- **Dictionary of the two mirrors** (`registrazioni/premessa_45/DIZIONARIO_DUE_SPECCHI.md`): map G, the geometry of the reading between the visible and the hidden sector (√3 per direction: lengths √3, surfaces 3, volumes 3√3; our own mass is not rescaled); map A of MM 2.44, the internal units of the hidden sector (A ≳ 1000). If the two turned out to be the same conversion of lengths, this dictionary would fail.

3. The galaxy scale as $cH_0/6$ (`codice/03_two_mirror_scale.py`)

- (cH_0/2)/3 = $1.091\cdot10^{-10}$ (H_0 = 67.4; −0.041 dex from 1.20) or $1.182\cdot10^{-10}$ (H_0 = 73.0; −0.007 dex); both inside the SPARC systematic band 0.96-$1.44\cdot10^{-10}$.
- **How it was found:** the factor was counted as «two directions» knowing that it gives the right number. In the family (cH_0/2)/3^(p/2) only p = 2 lands within 0.1 dex; since the family spacing (0.239 dex) is wider than that window, at most one member can fall in it, and that uniqueness is not evidence. The interest of the reading is coherence: the same rule on three directions is the 3√3 of MM 2.44. The coincidence with Verlinde concerns the coefficient only.
- **Coherence checks** (approximate estimates):
- near the Sun, with the baryonic speed that gives a rotation of 229 km s^{-1} for each scale (so the rotation is imposed, not tested), the vertical surface density within 1.1 kpc is 63.9 M⊙ pc^{-2} with $cH_0/6$ and 107.4 with cH_0/2, against 68 ± 4;
- the 3√3 cluster baryon fraction 1/(1 + 3√3) = 0.161 does not contain a_0 (unchanged by construction); in an example cluster (10^{14} M⊙ of baryons within 1 Mpc) the RAR mass drops by 3.9 %, so the known deficit grows slightly;
- evolution $a_0 \propto H(z)$ on the four MUSE-DARK III bins (errors symmetrised, no covariances): with free amplitude the shape is compatible if the bars are 1σ (χ^2 = 5.48 for 3 degrees of freedom, p = 0.14) and not if they are 95 % (χ^2 = 21.04, p ≈ 10^{-4}); with each of the three fixed normalisations considered ($cH_0/6$ with two H_0, and 1.20) the χ^2 are high, the bins lying on average 16-27 % above.
- **Not established:** that the field spreads over exactly two directions between the mirrors, the direction of the factor, the identification with a_0. $cH_0/6$ and $cH_0/2\pi$ differ by 4.7 % and cannot be separated today.

4. The threshold and readings that fall (`codice/01_threshold_and_mcm.py`, `codice/02_sparc_central_readings.py`)

- With counts — capacity C(r) = c²r/(2Gm), stacked cells N = M/m, U = M_U/m — the onset C(r)² = N·U gives

$r_{on}^2 = R_s(M) \cdot R_s(M_U)$ and, with $g = GM/r^2$ (spherical-equivalent mass), the threshold $g = c^4/(4G \cdot M_U) = c^2/(2R_s(M_U))$; the cell m cancels. With the true mass of a disc a geometric factor can reappear. Formally this is the surface gravity of a Schwarzschild mass M_U (an identity, not an identification).

- $M_U = M_H = c^2 R_H/(2G)$, the mass-energy of a critical-density sphere of radius R_H taken as the limit of the reading, gives $cH_0/2$. Reproducing a_0 requires $M_U = 2.728\ M_H$ (one quarter of Milgrom's «MOND mass» $c^4/(Ga_0)$); this is a required value, not a prediction, and does not establish a cosmological origin. Among the other boundaries examined (matter only, event horizon, particle horizon, de Sitter radius: 0.3-32 M_H) none is within 0.1 dex of 2.73.
- **Premise 43** tied the onset to the centre of each galaxy. Nine registered readings (the two-end reference and eight others) on 148 SPARC galaxies (quality $Q \leq 2$, inclination $\geq 30°$, at least five points, radii increasing, baryonic acceleration positive; central density from the first point, component masses as $v^2 r/G$): the reference gives a single scale 0.44 dex too high; readings with the central density give scales 70-2000 times too high; readings with the central potential or the constituents' readings (T1, T4, T5, T7) move the onset to lower accelerations but scatter by 0.79-0.99 dex; the «mcm» built from the masses of a galaxy's own constituents (maximum or sum, conventions of readings T8a and T8b) gives thresholds 13.3 and 13.2 dex too high; the scatter of the eight ranges from 0.26 to 0.99 dex. **None of the readings tried passes both registered criteria.** In the 63 galaxies where the onset can be located, its radius in disc lengths grows with the central disc density (Spearman +0.52), as a single acceleration scale implies.

5. Twelve X-COP clusters (`codice/04_xcop_clusters.py`)

- **Data and model:** hydrostatic total masses (NFW model and «forward» reconstruction, both assuming hydrostatic equilibrium and spherical symmetry), gas masses, stellar profiles for 7 clusters; for the other 5 the median star/gas ratio of those 7 is used. Two profiles end before 1.2 R500 (A3266 gas at 1.12, A2319 stars at 1.06 R500; last value held beyond); the preference for A/r persists in the two NFW fits truncated to the available coverage (A2319's extended stellar profile also enters the median star/gas ratio used for five clusters). Stellar profiles may be projected (≈ 2 % on a, from the earlier check, not recomputed by the module).
- **Registered test:** one scale per cluster at R500 against c200 and central entropy (at 0.03 R500; for A2255 and A2319 the entropy profile starts beyond that radius and the innermost bin is used): no indication (Spearman −0.09 and +0.02; +0.16 without those two). At 0.5 R500 the concentration correlates with the opposite sign to the one registered ($p = 0.006$; earlier check, not recomputed by the module).
- **Observed afterwards (declared):** with $a = g_{bar}/[\ln(1 - f_b)]^2$, $f_b = g_{bar}/g_{tot}$, the scale decreases with radius in all twelve clusters; median slope −0.97, range −1.44 to −0.61, between 0.2 and 1.2 R500.
- **A/r against a constant scale** (one amplitude per cluster, RMS of the residuals in log g_tot over 21 interpolated radii, not independent measurements): A/r better in 12/12; mean RMS 0.023 against 0.105 dex (NFW mass), 0.040 against 0.121 dex (forward mass). $r \cdot a(r)$ is constant only approximately: within a cluster it varies by up to ×2.2 (NFW) and ×6.5 (forward).
- **MM reading:** in galaxies $S = r/R_H$ grows point by point; in clusters S stays approximately fixed, $S = R_{fix}/R_H$, and $L/S \propto G \cdot M_{bar}(<r)/r$, a quantity with the dimensions of a potential (v_{bar}^2), not the full gravitational potential of an extended body. Then $R_{fix} = 2A/(cH_0)$: median 0.92 R500 (NFW) or 0.98 R500 (forward); with the factor 1/3 of section 3, $6A/(cH_0)$: median 1.82 R200c (NFW) or 1.83 R200c (forward), a scale compatible with splashback radii of massive clusters. These are medians of a conversion of A, not measured boundaries; splashback radii of these clusters are not measured here.
- **Sensitivity to the total mass** (version 2.46.1, `codice/06_mass_sensitivity.py`): for small baryon fractions $a \approx g_{tot}^2/g_{bar}$; at fixed radius and baryonic mass $d \ln a/d \ln M_{tot} = 2.06$-$2.27$ over the baryon fractions of the fitted X-COP nodes. At fixed radii, baryonic profiles and reference R500, increasing the X-COP total masses by 10 % increases the fitted R_fix by about 23 %; decreasing them by 10 % reduces it by about 20 %. Reconstructing A1795 and A2029 with the gas and NFW mass models of Vikhlinin et al. (2006), using the median X-COP stellar-to-gas ratio, gives $3R_{fix}/R200c = 3.3$ and 3.0 instead of 1.8 and 2.2 (a comparison between reconstructions: gas model and fitted range differ too; the V06 total mass of A1795 is 30-37 % higher at the three radii checked, 0.3, 0.6 and 1.0 Mpc). The inferred fixed-end radius depends strongly on the adopted mass profiles; its proximity to splashback scales is therefore calibration-dependent.
- **Ordinary explanation:** for small baryon fractions $d \ln a/d \ln r \approx m - 2 - \beta$, with $m = d \ln M_{tot}/d \ln r$ and $\beta = d \ln f_b/d \ln r$; NFW-like masses ($m \approx 1.2$) with rising baryon fractions ($\beta \approx 0.2$) give ≈ -1. The data do not distinguish this from the MM reading.
- **Other samples:** the constant scale inferred from a profile that decreases with radius is higher where the inner radii weigh more; this may explain part of the spread among CLASH, X-COP and groups (not all of it: A differs too).

6. CLASH cores (`codice/05_bcg_cores.py`)

- **Premise 47:** in a cluster the galaxies are far apart even if they look close in projection; in a galaxy the stars are close; the type of fractionation seen from the observer changes the observed behaviour. Registered test: inside

the brightest cluster galaxies the scale should read like galaxies.

- **Registered outcome: no indication.** In 10 of 13 cores the central values of the baryonic mass exceed the lensing mass (a tension between estimates, not a measurement of zero dark matter); the other three give a median $1.67 \cdot 10^{-10}$, just outside the registered band.
- **Exploratory, added afterwards** (simplified weighting: lensing error and 10 % on stars; gas errors and the propagation through the non-linear prediction not included): predicted over lensing mass is 2.0-6.4 (median 3.7) for the constant cluster scale, 5.0-15.3 (median 10.4) for A/r with the X-COP amplitude, 0.81-2.42 (median 1.53) for the galaxy law, 0.66-1.98 (median 1.20) for no dark part. Among the four prescriptions compared, no dark part has the lowest score (χ^2 27, against 62, 1112, 11 709). With these masses and prescriptions the cluster extrapolations overestimate the cores; the galaxy scale is not singled out. Tian et al. report that the BCGs lie on the cluster relation in their combined fit; the discrepancy with this core-by-core comparison is not resolved here (masses, function or construction of the accelerations).

7. What is not claimed

- No derivation: $a_0 = cH_0/6$, the two directions, the fixed distal end and its location are readings of measured numbers, found after seeing them.
- No distinction from ordinary explanations of the cluster profiles, and no observational confirmation of MM.
- No preference for $cH_0/6$ over $cH_0/2\pi$ or over a constant $1.20 \cdot 10^{-10}$.
- The cluster mass deficit of RAR-type laws is not solved; the cores' scale is not measured.

8. How this can turn out wrong

Four tests with decision zones (agreement, intermediate, failure), written before the analysis of the samples they name: `registrazioni/REGISTRATION_2.46_EN.md` (cluster slopes in new samples; fixed distal end against measured splashback radii; $a_0/(cH_0/6)$ with independent calibrations; cores at their measured accelerations). The published version of this record fixes the protocol before the analysis of the samples it names; later changes will be registered separately. In version 2.46.1 the registered R2 protocol and decision criteria are unchanged. Module 06 documents the sensitivity of R_fix to the adopted total-mass calibration; R2 will be evaluated according to the published criteria, and any additional sensitivity analyses will be reported separately. Any revised protocol will be dated and registered before the analysis to which it applies.

9. Contents

- `scarica_dati.py` (checks the archive, every file and the completeness against the manifest), `riproduci.py` (checks the local data against the manifest, then runs the modules), `codice/` (six modules and `comune.py`), `risultati/` (reference JSON and logs per module, `00_summary.json` with package versions). Python 3.11+ with numpy, scipy, sympy, astropy.
- `dati/`: the MUSE-DARK III bins, the CLASH table and, for module 06, the Vikhlinin et al. (2006) parameters of A1795 and A2029 (transcriptions of published numbers), `MANIFEST_SHA256.txt` of the third-party files fetched by `scarica_dati.py`.
- `registrazioni/premessa_43-premessa_48`: the author's words (Italian), translations and criteria registered before each computation, the original scripts and logs (with the paths of the author's machine) and the `REGISTRO.txt` of each folder; `REGISTRATION_2.46_EN.md`. The scripts of `premessa_48` are records of the work, not part of the 63 checks; they also read the arXiv sources of Vikhlinin et al. 2006, Gastaldello et al. 2007 and Sadhu & Tian 2024, which are not redistributed: `registrazioni/premessa_48/HOW_TO_RERUN.md` says how to fetch and check them.
- `documenti/note_di_verifica` (outcome notes, Italian; corrections in separate files), `documenti/rapporti_astra` (second assistant's reports, Italian), `LEGGIMI_IT.md`, `WHERE_THIS_COMES_FROM.md`, `fonti/SOURCES.md`.

10. How the record was made

The framework and its premises are the author's; the author is not a professional physicist. Translations, code and text were written with an AI assistant (Claude); checks by a second AI assistant (Codex) are documented in the reports kept, not by a human referee; new numbers of the second assistant were recomputed with code. Related records: MM 2.44 (DOI 10.5281/zenodo.23329080: $3\sqrt{3}$, $a_0 \propto H(z)$), MM 2.42.1 (DOI 10.5281/zenodo.23329094).

11. Changes

- **2.46.1** (13 October 2026): module 06 and the paragraph «Sensitivity to the total mass» of section 5; the note on R2 in section 8; `dati/vikhlinin2006`; working records of premise 48 with their notes and reports. Modules 01-05 and their reference results are those of 2.46.

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